

[Draft] Reflex- Detect Evaluation Proposal

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Author: Chao Wang

Reviewer: Gideon Kim (He Him) Rahul Chaudhary Karim Wahba Dafang He Chi Zhang

Status: [draft, in-review, approved]

Context

Reflex functions as a multi-agent, self-improving discovery framework (Source: [Reflex Vision](#)).

This proposal concentrates on the "Detect" stage of the Reflex pipeline, which involves the following components:

1. **Research Agent (PM Agent):** Continually scans the platform to identify potential areas for improvement.
2. **Analytics Agent (DS Agent):** Sizes hypotheses in relation to topline metrics, leverages VLM to validate actual pin content, and designs contradiction queries to rigorously test discoveries.
3. **Feedback Curator: (Self-improving Agent)** Converts human reviewer corrections into permanent system patterns to ensure past errors do not recur.

The intricate nature of this multi-agent framework necessitates a move beyond traditional, deterministic benchmarking; but quantifying system efficiency, reasoning coherence, and overall business impact is critical.

Goals

-  Investigate both internal and external evaluation methodologies, frameworks, and tooling to establish a comprehensive point of view on agent evaluation.
- **Develop a measurement framework** specifically for the "Detect" phase, tailored to PM Agent.
- **Define clear guidelines for the initial data collection process, human feedback collection and instrumentations.**
- **Improve Reflex with the end goal of realizing shipped experiments**

Non-Goals

- Establish a rigid, immutable evaluation framework; the framework must adapt and iterate alongside evolving system capabilities.
- Define measurement on DS agent as the integration of Agent-to-agent calls with Analytics Agent already comes up LLM-as-a-judge evaluation.
- Deliver evaluation frameworks for the subsequent "Build" or "Simulate" phases.

Evaluation Objective

1. Avoid Reactive Loop – catching issues before it reaches for human feedback.

- 2. Validate system metrics, including tool utilization, latency, and token consumption.
- 3. Verify reasoning trajectories by focusing assessment on outcomes rather than static knowledge.
- 4. Quantify and track concrete business outcomes to ensure Reflex delivers a measurable impact.
- 5. Optimize self-improvement cycles, drive ongoing product refinement, and uncover hidden insights

Methodologies

1. A hybrid grader type approach

Anthropic outlines [three types of graders for agents](#).

Code-based graders			Model-based graders			Human graders		
Methods	Strengths	Weaknesses	Methods	Strengths	Weaknesses	Methods	Strengths	Weaknesses
• String match checks (exact, regex, fuzzy, etc.) • Binary tests (fail-to-pass, pass-to-pass) • Static analysis (lint, type, security) • Outcome verification • Tool calls verification (tools used, parameters) • Transcript analysis (turns taken, token usage)	• Fast • Cheap • Objective • Reproducible • Easy to debug • Verify specific conditions	• Brittle to valid variations that don't match expected patterns exactly • Lacking in nuance • Limited for evaluating some more subjective tasks	• Rubric-based scoring • Natural language assertions • Pairwise comparison • Reference-based evaluation • Multi-judge consensus	• Flexible • Scalable • Captures nuance • Handles open-ended tasks • Handles freeform output	• Non-deterministic • More expensive than code • Requires calibration with human graders for accuracy	• SME review • Crowdsourced judgment • Spot-check sampling • A/B testing • Inter-annotator agreement	• Gold standard quality • Matches expert user judgment • Used to calibrate model-based graders	• Expensive • Slow • Often requires access to human experts at scale

At Pinterest, code-based grader and model-based grader are already adopted in multiple agent applications ([learn more](#)). For example, the Analytic Agent uses a pre-defined questions bank and rubrics to test the agent output quality, the test runs can be traced [on this dashboard](#). Model-based grader would be a good fit for a Data Analytics agent.

The PM agent in Reflex acts similar to research agents that gather, synthesize and create hypothesis cards. The output is non-deterministic and human feedback is considered as the strongest signal and gold standard. In such cases, the industry practice is to use human graders, or use LLM-as-a-judge, and use Human to calibrate the LLM judgement.

Given the subjective nature of research quality and evolving business strategies, **LLM-based rubrics should be frequently calibrated against human judgement to grade PM agent effectively.**

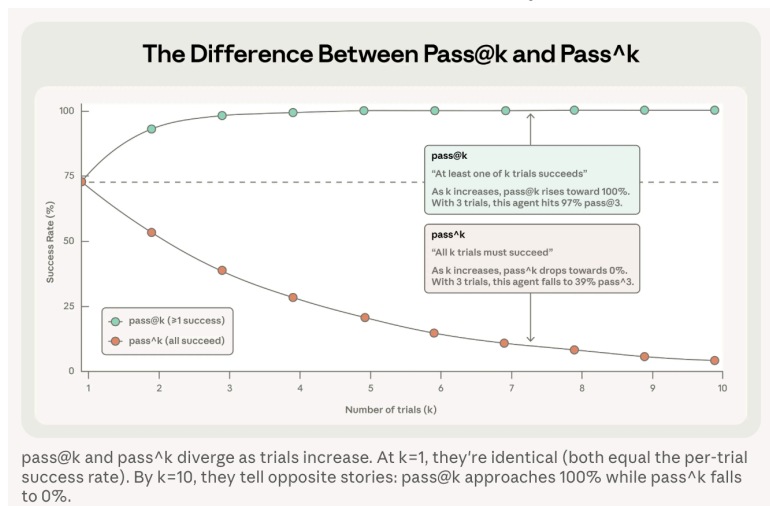
A hybrid grading approach is recommended [source](#):

- Utilize code-based graders to ensure comprehensive coverage, proper routing, and traceability and performance.(Task performance)
- Employ LLM-as-a-judge with human calibration to guarantee source quality and groundedness through cost-efficient grading.
- Incorporate human feedback and a human-curated gold dataset to safeguard recall and establish the benchmark for expected agent outputs.

2. Multi-trial Approach

Agent behavior varies between runs. Two metrics were recommended to cover the non-determinism and path variability.

- Pass @ K measures the likelihood that an agent gets at least one correct solution in k attempts.
- Pass ^ K measures the probability that all k trials succeed.



3. Building a gold standard dataset.

Developing a gold standard requires 20–50 reference examples. This involves first defining what constitutes a high-quality hypothesis card, an optimized data query, and reliable expert judgments, which relies on domain expert input. Three potential methods for gathering this human feedback have been identified:

- Utilizing historical PM roadmaps or brainstorming documentation, where past roadmap projects serve as the primary source of hypotheses. TODO: reach out to PM for high quality roadmap doc, reach out to DS for high quality opportunity sizing doc.
- Collecting human feedback directly on existing hypothesis cards, which necessitates removing all points of friction from the feedback submission process. TODO: tag/email domain expert on existing hypotheses.
- Humans generated Hypotheses cards with details (not rough ideas).

Proposal

Phase 0: Task Performance (Gideon Kim (He Him))

Metric	Definition	Objective	Grader Type
Phase completion rate	# of completed required phases / # of total required phases	Can be measured per agent (pm_agent, ds_agent) or end-to-end (detect cycle). A clear checklist is needed to trace the performance.	Code
Token Usage	Total Token (input+output) consumed per run	Cost efficiency	Code
Cost per Run	Dollar Cost of one full Detect cycle run	Performance tracing and future optimization	Code
Tool Selection Accuracy	# of correct tools selected/ # of total tools selected. Requires ground truth	Trajectory efficiency: Accuracy, oversearch, unnecessary tool use.	LLM as a judge with Human calibration
MCP count	# of unique MCP used per detect cycle	Trajectory efficiency: tool hopping.	Code
Total tool calls	# of all MCP tool invocations	Trajectory efficiency: looping/redundancy	Code

Phase 1: PM agent Reasoning (Chao Wang)

Metric (arranged by order of priority)	Definition	Objective	Grader Type
Precision of PM agent output	# of valid/ # of total generated hypothesis cards, relies	Correct interpretation, grounded hypothesis formed. A better metric would be measuring the unique cards rather than the raw cards. It is best to have at least one hypothesis rated based on playbooks.	LLM-judge with Human Feedback Calibrations
Hypothesis Survival To Opportunities	# of hypotheses card moved to opportunities/ # of total hypothesis card	Hypotheses are logical and grounded	Code
Playbook Coverage Rate	# playbooks with ≥ 1 valid hypothesis/ #total playbooks	Ensure agents have a broad coverage and balanced idea generated from a variety of playbooks. Entropy or max playbook concentration can also be used	Code
Hypothesis Merge Rate	K merged (unique) cards/ N agent created card	Reduce duplicative/repetitive ideas, avoid infinite loop	Code

Recall of Hypothesis	# gold hypotheses matched by PM agent/# total gold hypotheses, requires human curated hypotheses from past research, brainstorm sessions or roadmaps.	Missed product opportunities from playbook	Human (require PM domain expertise)
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Phase 2: Feedback curator & Skeptic Self Learning Capability

Metric	Definition	Objective	Grader Type
Learning Precision	# of high-quality learnings (rated by human)/ # of total learnings	Verify self-learning accuracy, usefulness.	Human
Learning Capture Coverage	#feedback comments converted into learning records/ #human feedback comments per run	Ensure agent captures learning from humans feedback	Code/LLM-as-a-judge
Artifact Routing Accuracy	#correctly categorized learnings / #total learnings	Ensure learnings are categorized to the right file	LLM-as-a-judge With human-calibration
Other metrics	Learning Reuse Rate, Repeat Mistake Rate	Ensure reusability.	

Phase 3: Business Impact

Metric	Definition	Objective	Grader Type
Human Usage Report (per eval cycle)	# of unique users engaged with cards # of unique cards engaged - Diversity of user roles - User churn Similar to Analytics Agent - User Behavior Analysis	Whether Reflex is integrated into the EPD workflow.	Code
Funnel Survival Rate to Shipped Experiment	Stage-by-stage card survival rate = # of cards in current stage / # of cards in the preceding stage *100%	Measure the relevance and feasibility of the hypothesis cards.	Code
Number of Shipped Experiment	Successful experiment launched into production.	Measure the incremental business impact of reflex	Code
Surface Goal Metrics; Topline Metric lift (SSV2, GPS 2.0)	Top line metric lift in surface holdout experiment from Reflex-proposed experiments	Measure the incremental business impact of reflex	Code

Currently not prioritized

- Safety, ethics, and compliance.
- Task latency: from input to output. Currently the agent is manual-triggered
- DS agent tool efficiency

- Human Hours Saved.

Deliverables

- Tracing/observability logs.
- Human Grader Gold standard dataset
- Eval agent repo Dashboard

Implementation Plan

- Data tracing, collection, code-based grader infrastructure
- Metric Refinement
- Ground truth from human input
- Hypothesis card
- Eval Visualization/Visibility
- Eval Iteration

Resources/References

Pinterest Internal Agent Evaluation Practice

1. [EvalHub](#): An industrial-scale offline evaluation platform for AI agents” (register agents, upload datasets, run simulations, score results with LLM or code graders). Repo: [pintools](#)
2. Helix Analytic Agent Evaluation: The dashboard visualizes the test results from a human pre-defined questions bank. [Repo](#), The [Analytics Agent - User Behavior Analysis - Apr 2026](#) summarized usage. [Analytics Agent Testing Process](#)
3. [Pinterest Dash](#) is evaluated based on three dimensions: [Dash Roadmap Q3 2026](#)
 - a. Final Answer Accuracy
 - b. Context Gathering
 - c. Workflow Decomposition

In discussion with [Rahul Chaudhary](#) mentioned the precisions are easier to evaluate either by human or LLM-as-a-Judge, but the recalls of what agent missed is harder to measure.
4. Ads Data& Reporting Agent is evaluated based on a pre-defined user queries that aims to cover all kinds of requests the agent expect to receive, [Ads Data & Reporting AI Agents](#)
5. [Ads Agent Eval Framework](#) employs A/B testing of prompts and LLM configs, using Langfuse
6. [Data Analysis with Langfuse](#). It outlines the type of metrics to track:
 - a. LLM usage metrics
 - b. Agent Behavior

- c. User Interaction
 - d. More advanced analysis approaches including A/B testing, user segmentation, session analysis
7. Unified-VLM-User-Bot-Simulation uses a [jupyter notebook](#) to track the cost per run

External Resources/References

[Agent Evaluation: How to Test and Measure Agentic AI Performance](#)

<https://langfuse.com/> Langfuse helps with the tracing, observing and improvement AI agents

<https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents>

<https://aws.amazon.com/cn/blogs/china/agent-quality-evaluation/>

<https://developer.nvidia.com/blog/mastering-agentic-techniques-ai-agent-evaluation/>

Phase 0: Task Performance Evaluation

Author: Gideon Kim (He Him)

Contributor: Tim Chu Chao Wang Dafang He

Last updated: Jul 16, 2026

Meeting notes Jul 21, 2026

[Gideon]

1. Tool selection is not covered. Maybe able to use some LLM-judge. Everything else is being tracked.

2. Taking the metrics we track, we use the audit stage. Right now doing an audit live. Caching This week:

Improve Audit; dashboard improve.

Todo: show chao how to get into the S3 bucket. Show

[Chao]

Meeting notes Jul 15, 2026

[Chao]

Understand your expectation walking out of this project. Understand your PRs, your collaboration with the Team.

[Gideon]

- Webapp dashboard (personal)
- Improve cycle_log.jsonl add four different updates, cost and token tracking, Syncing to S3. learning judgement.
- Google form script to expert leaving reviews – we don't actually need people to be the bottle neck.
 - Relevance, feasibility, impact. Pretty subjective.
 - Divided the hypotheses in different sections. Sectors of the company. For each subset of questions.
 - Have not figure out the domain expert.

Would like you to get your help on the Task Performance aspect.

Let's define some milestones and timeline.

Next step: research existing task performance observability approach used in Pinterest. Come up with an implementation plan.

Recap:

This week: Gideon Kim (He Him) focus on Task performance, share his current work on human feedback gathering tools/rubrics w/ Chao

Chao: Implement LLM as a judge on PM agent and playbooks, use findings to 1) identify low hanging fruit to improve the task processes, 2) draft the hypothesis rubrics for human feedback

Next week: regroup on Tuesday Jul 21, 2026 , discuss next steps.

Phase 1- PM Evaluation Plan: LLM Judge With Human Calibration

Date: Jul 23, 2026
Author: Chao Wang
Contributor: Dafang He Tim Chu Karim Wahba Gideon Kim (He Him)
Consultant: Raghav Jindal (GEPA)
Status: Version 1 of the LLM judge is built. [PR#63](#)

TL;DR

This work is focused on evaluating the quality of hypothesis cards generated by the PM agent in Reflex.

The broader plan has two stages:

- Stage 1: Optimize the LLM judge using human calibration.
- Stage 2: Use the LLM judge results to optimize the playbook prompts that generate the hypothesis cards.

The main reason for human calibration is simple: we do not yet know whether the current rubric is good enough, or whether the LLM judge is grading well enough. To address this, I plan to collect around 20 human-graded cards from domain PMs and use those results to improve the judgment through GEPA-based prompt optimization.

Once the judge becomes more reliable, I want to run it automatically in the detect cycle and use its outputs to improve upstream playbooks.

Two-Stage Evaluation Overview

Stage Name	Evaluation Objective	Methodology	Expected Evaluation Results	Simple Process View	Ground truth & availability
Stage 1: Optimize the LLM-Judge with Human Calibration	Improve the quality and reliability of the LLM-judge for scoring hypothesis cards	Create V0 of the LLM judge with a 5-dimension rubrics; collect ~ 20 human-graded hypothesis cards from domain PMs; compare LLM judge scores and rationales with human ratings; use GEPA to optimize the LLM-judge prompt based on calibration results.	A better-calibrated LLM judge with improved alignment to human judgment, more reliable dimension scores and composite scores, and a smaller gap between LLM and human grading	PM agent -> hypothesis card -> LLM judge v1 -> human calibration -> GEPA -> improved LLM judge.	Evaluation output from human graders. Limited in volume given the significant resources required for human-in-the-loop calibration.

Stage 2: Use LLM-Judge Results to Optimize Playbooks	Improve/optimize the upstream playbook prompts from the quality grading of hypothesis cards.	Run the calibrated LLM judge in the detect cycle at scale; score more hypothesis cards across playbooks; connect card quality back to the playbook used to generate each card; use GEPA to optimize playbook prompts based on judge results and score patterns.	Better playbook prompts, stronger hypothesis cards, and higher downstream success rates such as more cards passing to the opportunity stage	PM agent playbook -> hypothesis card -> calibrated LLM judge -> score patterns -> GEPA -> improved playbook prompt.	Shipped experiment. Very limited in volume, the funnel could introduce agnostic factors.
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The `PM_agent.md` uses playbooks to generate product improvement hypotheses. My goal is to evaluate how strong those hypothesis cards are and create a repeatable way to improve both:

- the evaluation system (stage 1)
- the generation system (stage 2)

Stage 1: Hypothesis Judge Optimization

The current LLM judge output includes: [example](#)

- Five dimension scores and rationale
- A composite score
- Overall verdict
- Strength, improvements, confidence

The methodology is:

1. Judge creation: Create V0 / V1 of the LLM judge
2. Human calibration:Collect ~20 human-rated hypothesis cards
3. Judge optimization:Compare LLM judge outputs with human ratings. Use GEPA to optimize the judge prompt
4. Judge optimization:Iterate until judge-human alignment improves

Stage 2: Hypothesis Card Evaluation, Playbook Optimization

The methodology is:

5. System optimization: Operationalize the judge in the detect cycle
6. System optimization: Use judge results and GEPA to optimize the upstream playbooks

Implementation

Human Feedback Collection Plan

1. Who: The current plan is to gather human feedback from domain expert product managers (PM). These are PMs who work in the same domain as the playbook used by the PM agent.
2. Why: I chose this approach because hypothesis quality is domain-sensitive, and domain PMs are the best reviewers for whether a card is actually useful.
3. How: Asana forms ([link](#)).
This approach has several benefits:
 - Human Accessibility: Lower friction for reviewers: PMs can respond through email without requesting Asana access
 - Data storage: The feedback is structured because the survey result automatically becomes an Asana task
 - Agent Accessibility: The results can later be retrieved through the Asana REST API
4. **Implementation Constraints**: I cannot automatically customize the form for each reviewer with the specific hypothesis card context. Possible workarounds:
 - a. Manually copy-paste card content to form, email the Human grader
 - b. Use HTML to display the content, attach the HTML to form invitation email, ask grader to copy-paste the HTML to survey result to keep track of card GID <> human grading results.

Sample Hypothesis cards

1. Alignment on the [open questions](#).
2. Short list PM human graders and matching hypothesis card.

LLM judge Calibration Implementation

Use GEPA for prompt optimization. [DSPy](#) package available.

Risks and Open Questions

1. No Strong Ground Truth of Hypothesis Card

The biggest risk is that we do not yet have a strong ground truth for hypothesis quality. They are slow to gather and may be affected by confounding variables in the reflex funnel.

Possible mitigations: Current Proxy Metrics

- 1) Whether a hypothesis card can pass to the opportunity stage
- 2) Human Graded Scores

2. No Ground Truth for LLM judge quality

With few human-labeled examples, it is hard to prove absolute quality well.

Possible mitigations:

Track whether the judge is stable, aligned, and not regressing.

3. Lack of Inter-rater consistency

The current plan is to gather ~20 human feedback. This creates a risk because human graders are also subjective.

A common industry practice is to have multiple human graders review the same hypothesis card so that disagreement can be measured and calibration can be more reliable.

4. Human Feedback May Not Be Useful Enough

One concern is that multi-dimensional scoring may be harder to optimize against than a simpler label. Netflix's approach reduced evaluation to a binary outcome such as pass / non-pass. My current rubric has five dimensions, which may create more complexity.

Possible mitigations:

- Mandatory written rationale from human reviewers
- Add a binary question such as "is this hypothesis pass or no pass"
- Consider pairwise ranking between two hypothesis cards, possibly use stack ranking or ELO-style scoring later

5. Uneven Coverage Across Playbooks

Some playbooks may receive direct human feedback, while other playbooks may have little or no labeled data. This creates a risk for prompt optimization. Playbooks with direct human feedback may improve faster, while other playbooks may only learn indirectly from patterns observed in different domains or subjects. How much cross-playbook transfer should we allow, and do we need a minimum amount of direct human feedback for each playbook before using optimization results?

Open question, discuss live on Jul 24, 2026

1. [Decision: Both] Human grade PM-agent strengthened card or playbook-based card? Our goal is to improve the playbooks.
2. [Decision:] Should we require multiple human PM graders for at least a subset of hypothesis cards so we can measure inter-rater consistency?
 - a. potential
3. [Feedback:] Need your feedback on the 5 dimension rubric
 - a. Comprehensive but we can ask humans to rate a more general aspect, the actionability can be left to later stages.
 - b. Eng may be the better human expert to grade the actionability of the proposal.

4. [Decision:] Threshold: when is LLM-judge good enough?

Reference

GEPA: Reflective Prompt Evolution for Annotation Generation

<https://github.com/pinternal-dev/annotations-eval/tree/main>

<https://netflixtechblog.com/evaluating-netflix-show-synopses-with-llm-as-a-judge-6269251e6f28>

Meeting Recap:

1. Aligned on loosening on technical solution proposals, DS agents should handle it. We should allow some core problem themes (e.g. relevance improvement), those one may not be able to solve with single solutions
2. Humans should grade both playbook-generated cards and PM agent strengthened cards.
3. Asana access should not be a blocker to gather human PM feedback.
4. James: use binary pass/no pass might be better, afraid llm judge may get lost in such comprehensive ratings.
5. Tim: want to see all LLM judge do without human calibration — chao: likely need to use HTML to better display the logs, and also directly gather human feedback from it.
6. James: make sure to curate and hold out a strong, high-quality golden dataset if using GEPA, be careful with reward-hacking.
7. Sample size of human feedback? 20, run as a pilot program. Continuous Eval or one time?
8. Gideon/Tim: loop in with Janvi's build agent eval.

Next steps: Jul 24, 2026 1-2 weeks

1. Tune down the weight of actionability within the five dimensions.
2. Run LLM judge without human feedback, visualize/share results.
3. Identify Human PM grader, match with Hypothesis cards. Add binary questions to Asana form, gather feedback.
4. Use GEPA to prompt optimize the LLM judge.